Sri Lanka Institute of Information Technology

Data Warehousing and Business Intelligence

IT3021

Assignment 2

IT3021 - Year 3, Semester 2

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01 Data source for the assignment – 2

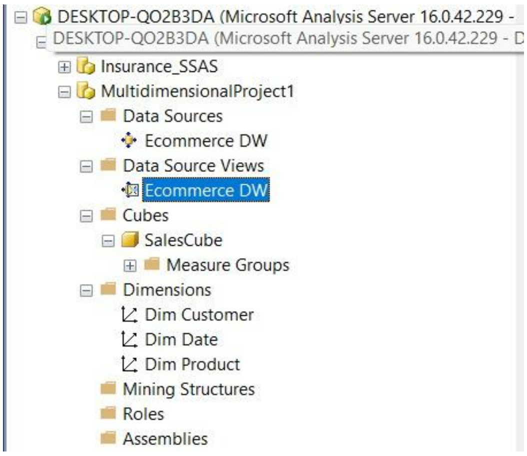
1.1 Data Source Introduction

The data warehouse used for this assignment was created using an **E-Commerce Sales Transactions Dataset**. This dataset contains transactional data related to customer purchases in an online retail environment. The dataset includes detailed information about orders, customers, products, and sales performance.

The dataset consists of attributes such as **OrderID, CustomerID, ProductID, OrderDate, ShipDate, Region, Category, SubCategory, Quantity, Discount, SalesAmount, and Profit**. These attributes were transformed and loaded into a structured data warehouse named **ECommerce_Sales_DW**.

The dataset supports business analysis such as identifying profitable products, analyzing customer purchasing behavior, and evaluating sales trends across different regions and time periods.

Figer-1



1.2 Data Source Introduction

1. **DimCustomer** – Contains customer-related details such as name, segment, and location.
2. **DimProduct** – Contains product-related details such as category and subcategory.
3. **DimDate** – Contains time-related information such as year, month, and quarter.

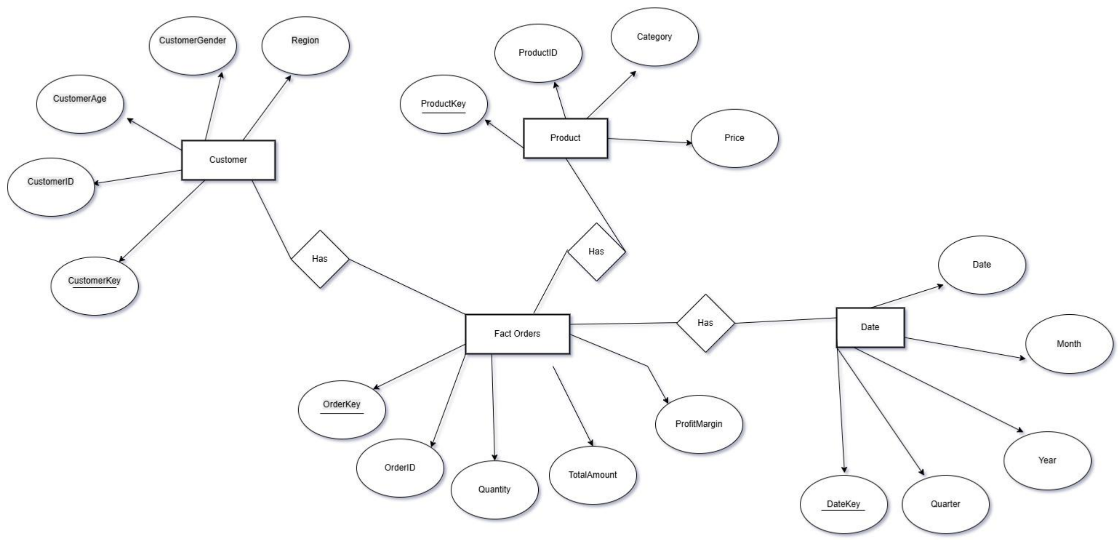
Fact Table

FactSales – Contains transactional data including:

- Quantity
- Discount
- SalesAmount
- Profit

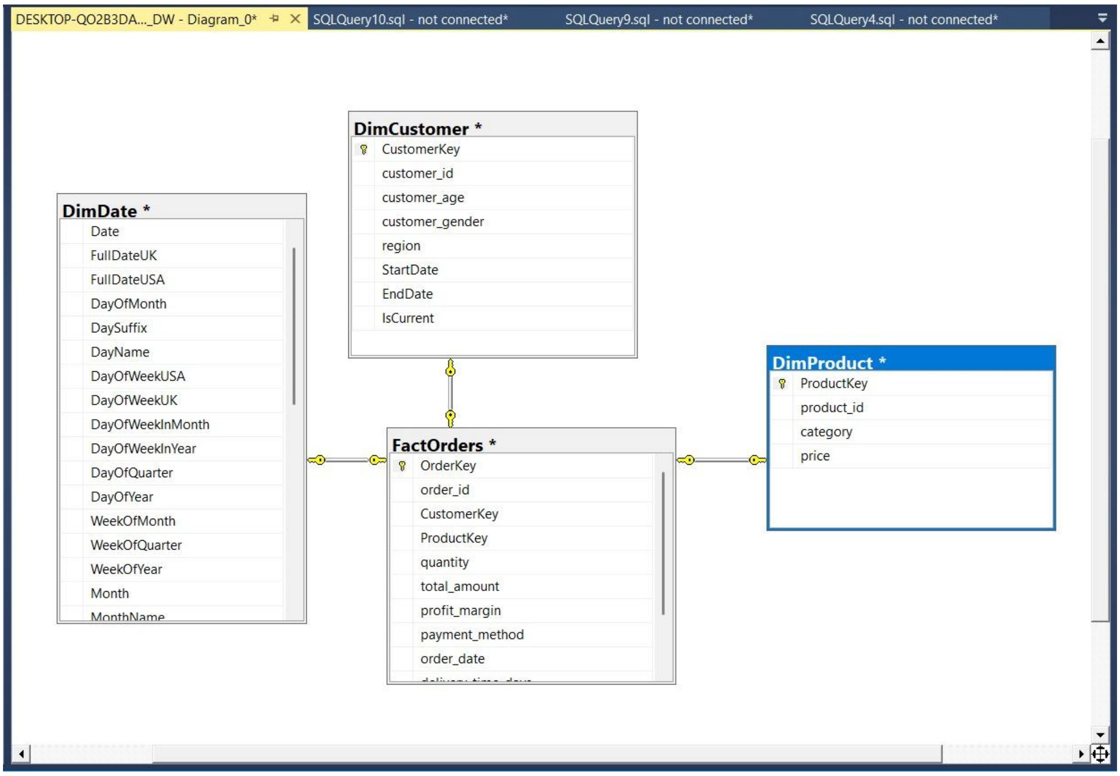
This table connects to all dimension tables through foreign keys.

Data warehouse ER



Figer-2

Implemented DW



Figer-3

02 SSAS Cube implementation

2.1 Cube Implementation

An OLAP cube was implemented using **SQL Server Analysis Services (SSAS)** to enable fast and efficient multidimensional analysis.

The cube consists of:

- **Dimensions** – Customer, Product, Date
- **Measure Group** – FactSales

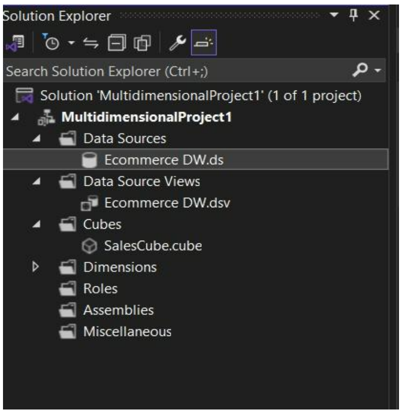
2.1.1Creating the Data Source

The data source was created by connecting to the **ECommerce_Sales_DW** database using SQL Server Data Tools.

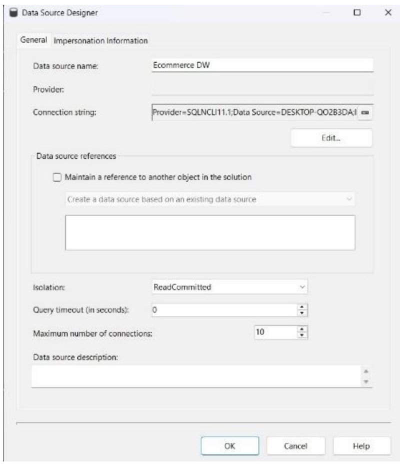
Steps included:

- Creating a new Analysis Services project
- Connecting to SQL Server
- Selecting the data warehouse database

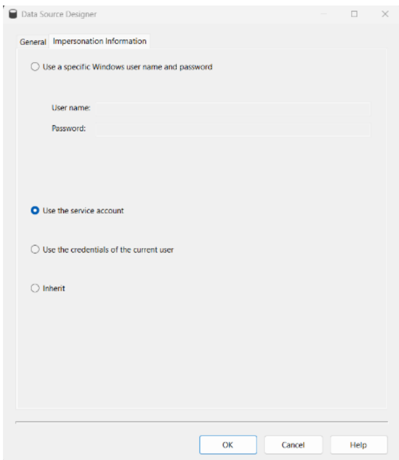
Solution Explorer



Figer-4



Figer-5



Figer-6

2.1.2 Creating the Data Source View

The Data Source View (DSV) was created to define the logical structure of the data.

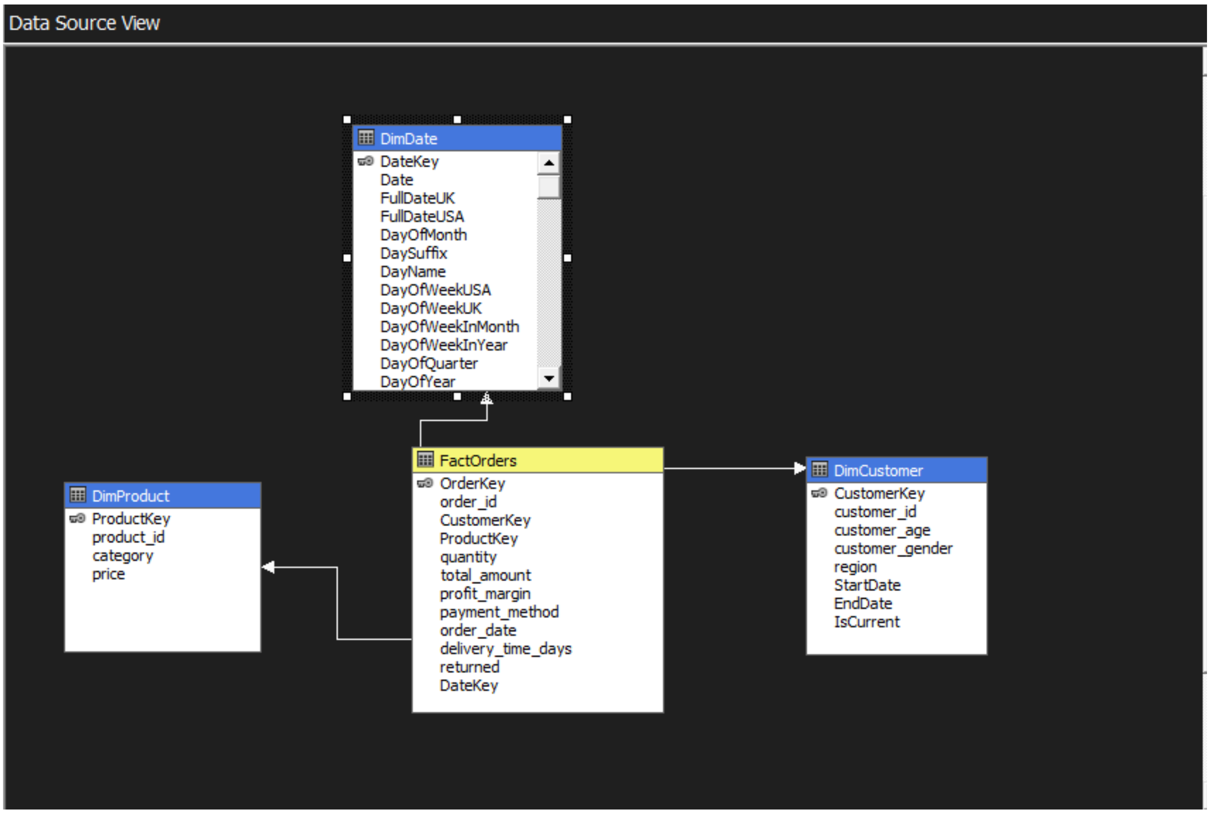
Tables included:

- DimCustomer
- DimProduct
- DimDate
- FactSales

Relationships between tables were verified.

.

Data Source View



Figer-7

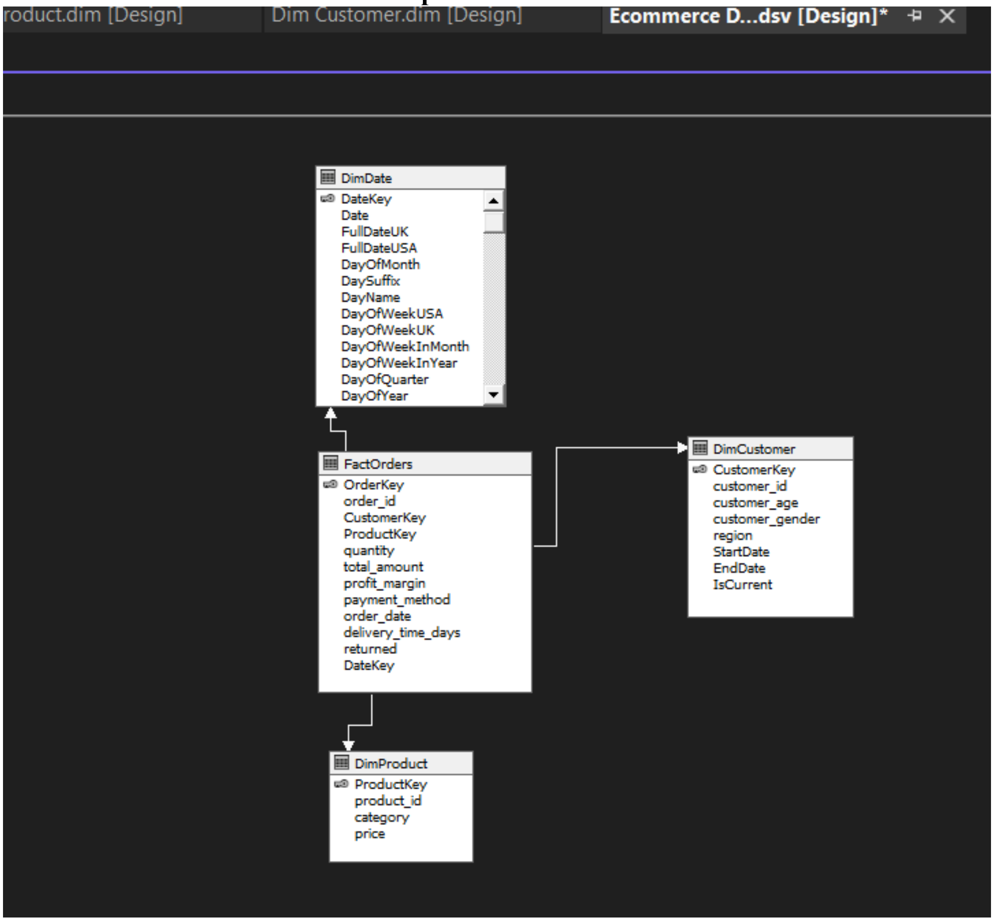
2.1.3 Creating the Cube

The cube was created using the Cube Wizard.

Steps:

- Selected FactSales as the measure group
- Selected measures: SalesAmount, Profit, Quantity, Discount
- Selected dimension

Implemented Cube



Figer-8

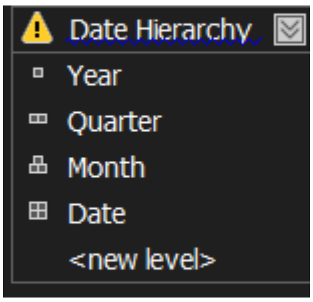
2.1.4 Creating Hierarchies and Dimension Structures

Hierarchies were created within dimensions to support drill-down and roll-up analysis. For example, the date dimension includes a hierarchy of Year, Quarter, Month, and Day. Similarly, the product dimension includes Category and Subcategory levels. These hierarchies make it easier for users to navigate data and analyze it at different levels of detail. By using hierarchies, users can start with a high-level overview and drill down into more detailed information.

Date Hierarchy

- Year → Quarter → Month → Date

Figer-9



Customer Hierarchy

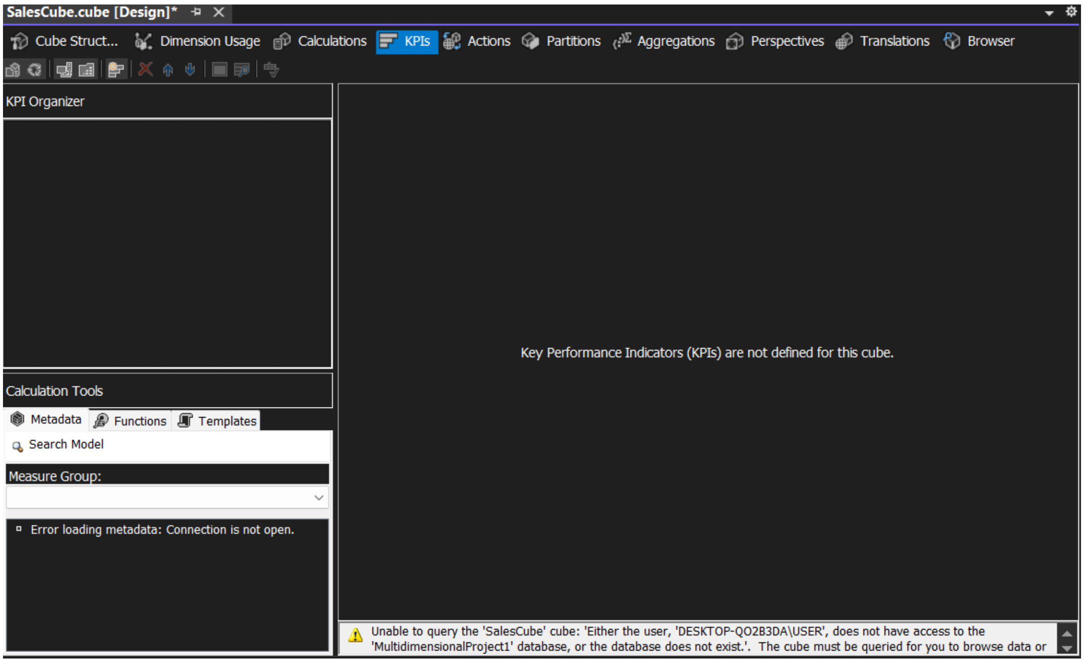
- Country → Region → State → City

Product Hierarchy

- Category → SubCategory → Product

2.1.5 Creating KPIs

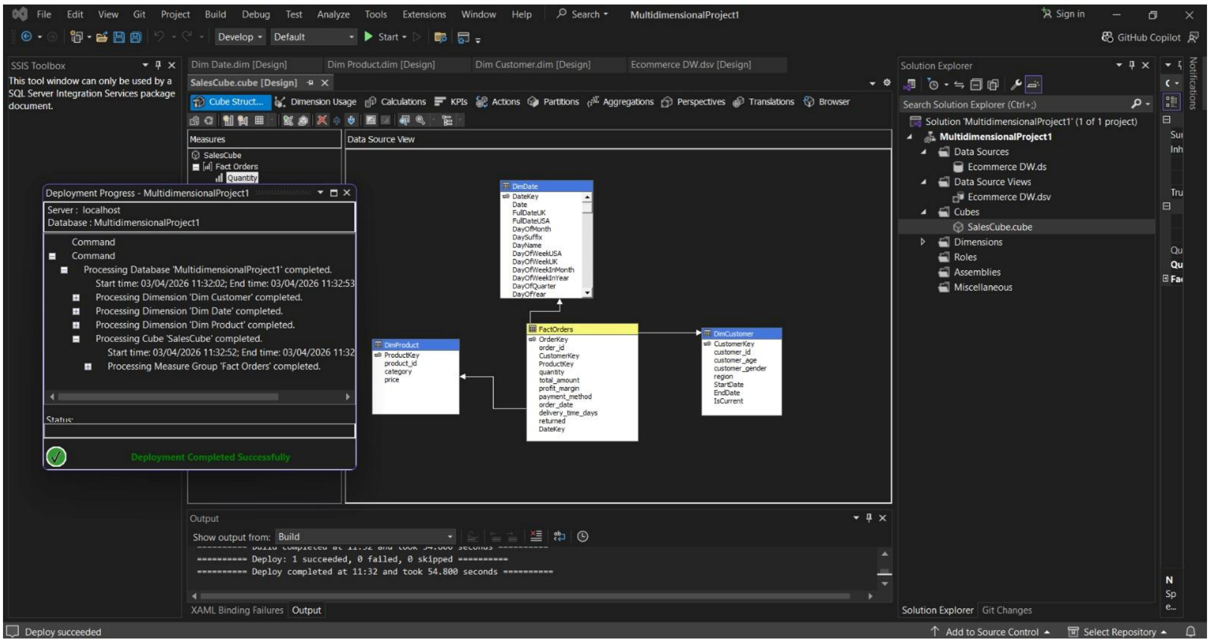
Key Performance Indicators (KPIs) were created to measure business performance. KPIs such as Total Sales, Profit Margin, and Order Count were defined in the cube. These indicators help users quickly understand whether the business is performing well or not. KPIs provide visual cues and benchmarks, making it easier to identify trends and performance issues.



Figer-10

2.1.6 Deploying the Cube

Finally, after all the above was done, the finalized cube was deployed



Figer-

03 Demonstration of OLAP Operations

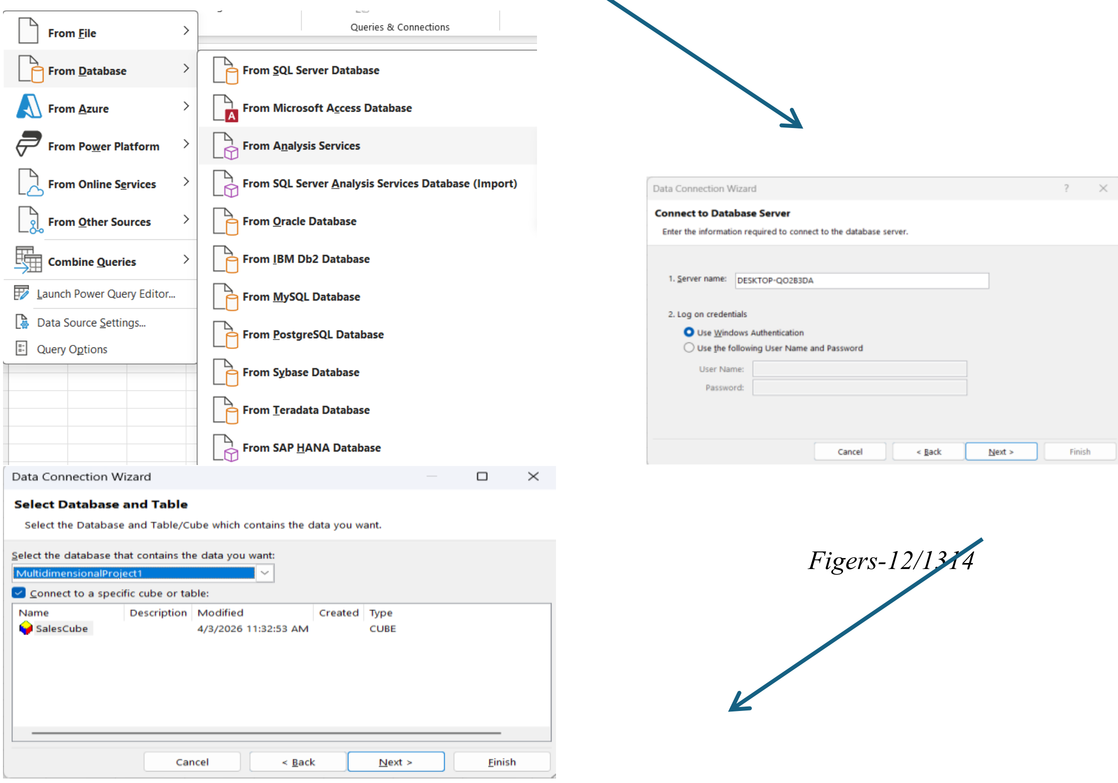
OLAP (Online Analytical Processing) operations are performed to demonstrate how data can be analyzed from multiple perspectives. These operations allow users to explore data interactively and gain meaningful insights. They are essential for multidimensional analysis and effective decision-making in business environments.

OLAP is an important component of Business Intelligence (BI), as it supports trend analysis and various types of data analysis from different viewpoints. It enables easy understanding and efficient handling of large amounts of data when making important business decisions.

Main OLAP Operations:

1. **Drill Down**
   Drill down converts less detailed (summarized) data into more detailed data. This is achieved by moving down the concept hierarchy.
2. **Roll Up**
   Roll up is the opposite of drill down. It aggregates data by moving up the concept hierarchy, resulting in more summarized information.
3. **Slice**
   Slice selects a single dimension from the OLAP cube and creates a smaller sub-cube based on that dimension.
4. **Dice**
   Dice selects a sub-cube by choosing two or more dimensions from the OLAP cube, allowing more specific data analysis.
5. **Pivot (Rotate)**
   Pivot rotates the data cube to provide a different view of the data, helping users
6. analyze it from another perspective

3.1 Connecting to the SSAS Cube



Figers-12/1314

3.2 OLAP Operations Demonstration Excel Report

3.2.1 Roll Up

Roll-up is used to aggregate data to a higher level. For example, sales data can be summarized from monthly to yearly totals. This operation helps users view overall trends and patterns without focusing on detailed data. It is useful for getting a quick overview of business performance.

				2023	2023	2023	2024	2024	2024	2025	2025	2025	Total Quant	Total Total Amou	Total Profit Marg
Customer Id	Category	Month	Price	Quantity	Total Amount	Profit Margin	Quantity	Total Amount	Profit Margin	Quantity	Total Amount	Profit Margin			
C10000	Electronics	9	177.46							4	709.84	56.6	4	709.84	56.6
C10000	Fashion	8	38.97				4	132.44	22.2				4	132.44	22.2
C10001	Electronics	3	696.25							12	21305.24	804.76	12	21305.24	804.76
C10001	Electronics	8	187.56				12	6752.12	231.88				12	6752.12	231.88
C10001	Home	1	225.62				12	8122.28	724.32				12	8122.28	724.32
C10001	Sports	11	31.60				4	101.12	1.48				4	101.12	1.48
C10001	Sports	3	102.88				8	1646.04	213				8	1646.04	213
C10002	Beauty	11	68.53	4	219.24	67.56							4	219.24	67.56
C10002	Beauty	8	22.70				4	99.48	11.8				4	99.48	11.8
C10002	Electronics	8	108.11				4	367.52	17.76				4	367.52	17.76
C10002	Home	1	42.45				8	272.48	32.12				8	272.48	32.12
C10002	Home	3	16.01				4	44.84	-3.76				4	44.84	-3.76
C10003	Grocery	6	1.08							4	4.08	-4.08	4	4.08	-4.08
C10003	Home	6	115.13				4	460.48	95.76				4	460.48	95.76
C10003	Home	7	38.14							4	152.52	13.44	4	152.52	13.44
C10004	Electronics	11	622.80	4	2491.16	254.96							4	2491.16	254.96
C10004	Home	12	59.78	4	239.08	42.04							4	239.08	42.04
C10004	Toys	1	11.47				12	412.88	33.76				12	412.88	33.76
C10005	Home	11	190.28				8	202.36	12.56				8	202.36	12.56

Figer-15

3.2.2 Drill Down

Drill-down is the opposite of roll-up. It allows users to view more detailed data by breaking down aggregated values. For example, users can analyze yearly sales data at the monthly or daily level. This operation helps in identifying specific patterns or issues within the data.

Row Labels	Quantity	Profit Margin	Total Amount
C10000	8	78.8	842.28
C10001	48	1975.44	37926.8
C10002	24	125.48	1003.56
C10003	12	105.12	617.08
C10004	20	330.76	3143.12
C10005			
Home			
11			
190.28	8	12.56	202.36
2			
107.31	4	93.08	429.2
Sports			
12			
24.43	4	75.72	334.6
C10006			
Electronics			
2			
71.37	4	10.44	228.36
Sports			
3			
149.32	8	325.2	2389.12
C10007			
Electronics			
12			
277.00	8	60.76	398.52
Fashion			

Pivot Table +

Figer-16

3.2.3 Slice

Slice is used to filter data based on a single dimension. For example, users can view sales data for a specific region or time period. This operation simplifies analysis by focusing on a particular subset of data.

Customer Id	Category	Month	Price	2023 Quantity	2023 Total Amount	2023 Profit Margin	2024 Quantity	2024 Total Amount	2024 Profit Margin	2025 Quantity	2025 Total Amount	2025 Profit Margin	Total Quant	Total Total Amou	Total Profit Marg
C10000	Electronics	9	177.46							4	709.84	56.6	4	709.84	56.6
C10000	Fashion	8	38.97				4	132.44	22.2				4	132.44	22.2
C10001	Electronics	3	696.25							12	21305.24	804.76	12	21305.24	804.76
C10001	Electronics	8	187.56				12	6752.12	231.88				12	6752.12	231.88
C10001	Home	1	225.62				12	8122.28	724.32				12	8122.28	724.32
C10001	Sports	11	31.60				4	101.12	1.48				4	101.12	1.48
C10001	Sports	3	102.88				8	1646.04	213				8	1646.04	213
C10002	Beauty	11	68.53	4	219.24	67.56							4	219.24	67.56
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C10004	Electronics	11	622.80	4	2491.16	254.96							4	2491.16	254.96
C10004	Home	12	59.78	4	239.08	42.04							4	239.08	42.04
C10004	Toys	1	11.47				12	412.88	33.76				12	412.88	33.76
C10005	Home	11	190.28				8	202.36	12.56				8	202.36	12.56

Figer-17

3.2.4 Dice

Dice is similar to slice but involves multiple dimensions. Users can filter data based on several conditions, such as region, category, and year. This provides a more detailed and focused analysis compared to slicing.

	Category	Month	Price	rgin	2024 Quantity	2024 Total Amount	2024 Profit Margin	2025 Quantity	2025 Total Amount	2025 Profit Margin	Total Quant	Total Total Amou	Total Profit Marg
4	Electronics	9	177.46					4	709.84	56.6	4	709.84	56.6
5	Fashion	8	38.97		4	132.44	22.2				4	132.44	22.2
6	Electronics	3	696.25					12	21305.24	804.76	12	21305.24	804.76
7	Electronics	8	187.56		12	6752.12	231.88				12	6752.12	231.88
8	Home	1	225.62		12	8122.28	724.32				12	8122.28	724.32
9	Sports	11	31.60		4	101.12	1.48				4	101.12	1.48
10	Sports	3	102.88		8	1646.04	213				8	1646.04	213
11	Beauty	11	68.53	67.56							4	219.24	67.56
12	Beauty	8	22.70		4	99.48	11.8				4	99.48	11.8
13	Electronics	8	108.11		4	367.52	17.76				4	367.52	17.76
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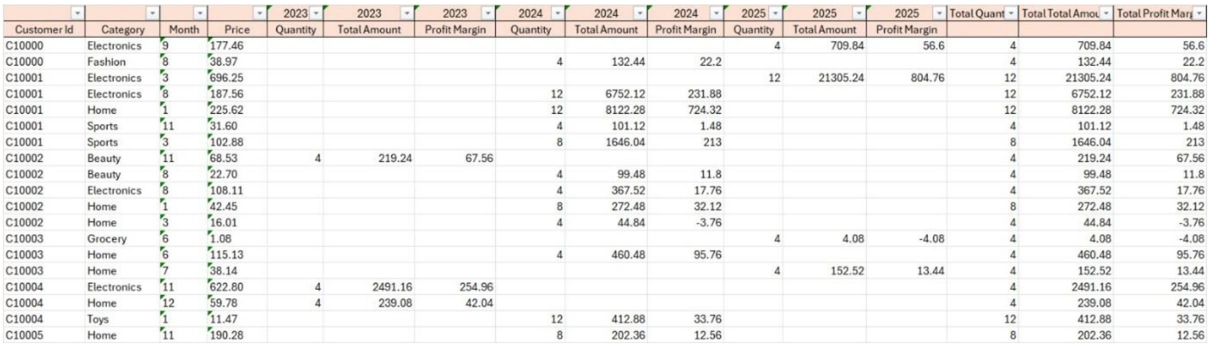
16	Grocery	6	1.08					4	4.08	-4.08	4	4.08	-4.08
17	Home	6	115.13		4	460.48	95.76				4	460.48	95.76
18	Home	7	38.14					4	152.52	13.44	4	152.52	13.44
19	Electronics	11	622.80	54.96							4	2491.16	254.96
20	Home	12	59.78	42.04							4	239.08	42.04
21	Toys	1	11.47		12	412.88	33.76				12	412.88	33.76
22	Home	11	190.28		8	202.36	12.56				8	202.36	12.56
23	Home	2	107.31					4	429.2	93.08	4	429.2	93.08
24	Sports	12	24.43	75.72							4	334.6	75.72
25	Electronics	2	71.37					4	228.36	10.44	4	228.36	10.44
26	Sports	3	149.32		8	2389.12	325.2				8	2389.12	325.2
27	Electronics	12	277.00		8	398.52	60.76				8	398.52	60.76

Region
Central
East
North
South
West
Unknown

Figer-18

3.2.5 Pivot

Pivot is used to change the orientation of data. It allows users to rearrange rows and columns to view data from different perspectives. This operation makes it easier to compare values and identify patterns.



Customer Id	Category	Month	Price	2023 Quantity	2023 Total Amount	2023 Profit Margin	2024 Quantity	2024 Total Amount	2024 Profit Margin	2025 Quantity	2025 Total Amount	2025 Profit Margin	Total Quant	Total Total Amou	Total Profit Marg
C10000	Electronics	9	177.46							4	709.84	56.6	4	709.84	56.6
C10000	Fashion	8	38.97				4	132.44	22.2				4	132.44	22.2
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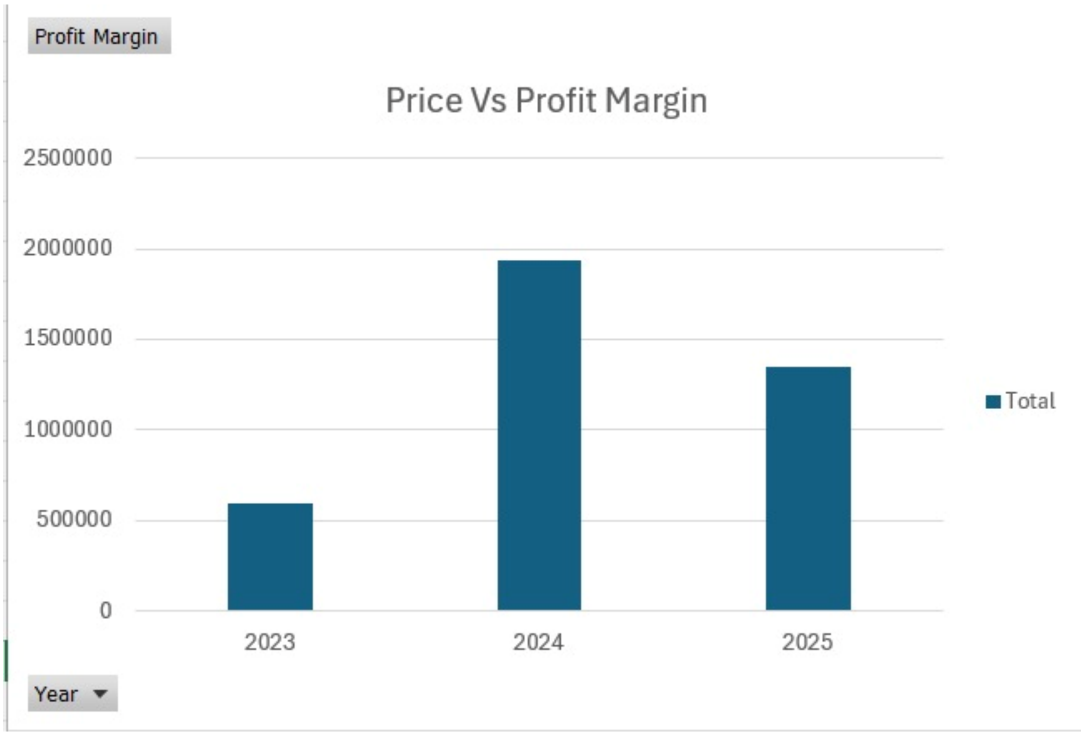
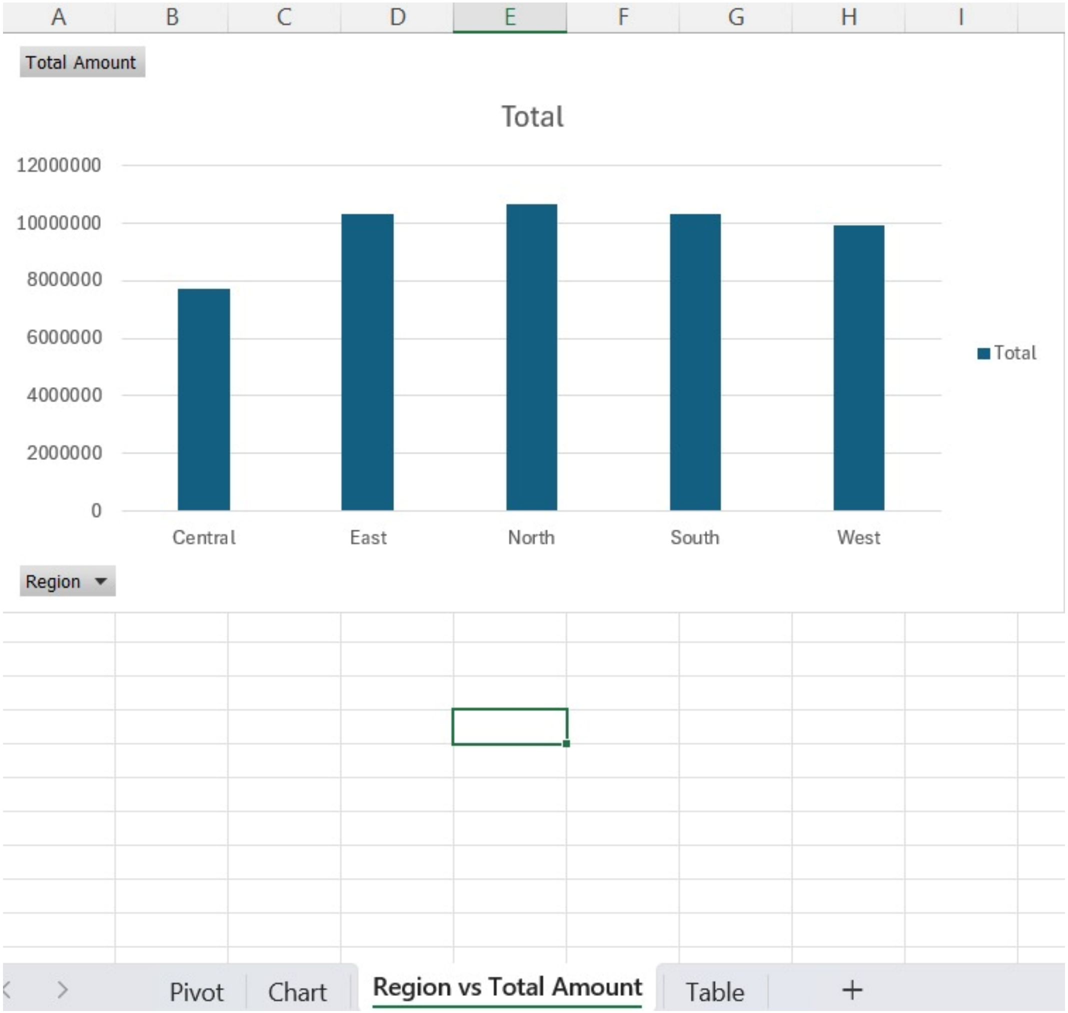
C10004	Toys	1	11.47				12	412.88	33.76				12	412.88	33.76
C10005	Home	11	190.28				8	202.36	12.56				8	202.36	12.56

Figer-19



Figer-20


A
B
C
D
E
F
G
H
I
Total Amount
Total
12000000
10000000
8000000
6000000
4000000
2000000
0
Central
East
North
South
West
Total
Region
Pivot
Chart
Region vs Total Amount
Table
+

Figer-21

04 Power BI Reports

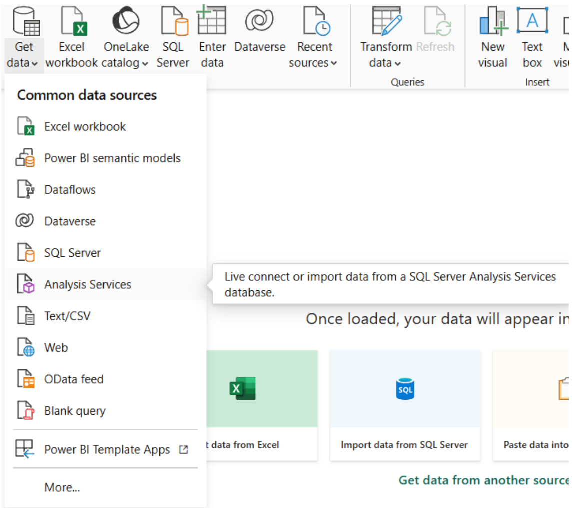
Power BI was used to create interactive reports and dashboards based on the SSAS cube. It is a powerful data visualization tool that allows users to present data in graphical formats such as charts, tables, and maps. By using Power BI, users can gain insights quickly and make data-driven decisions.

4.1 Connecting to the Data Source

Power BI was connected to the SSAS cube using the “Get Data” option. This connection allows real-time access to the cube data. Users can create reports directly from the cube without importing data, ensuring that the reports always display up-to-date information.

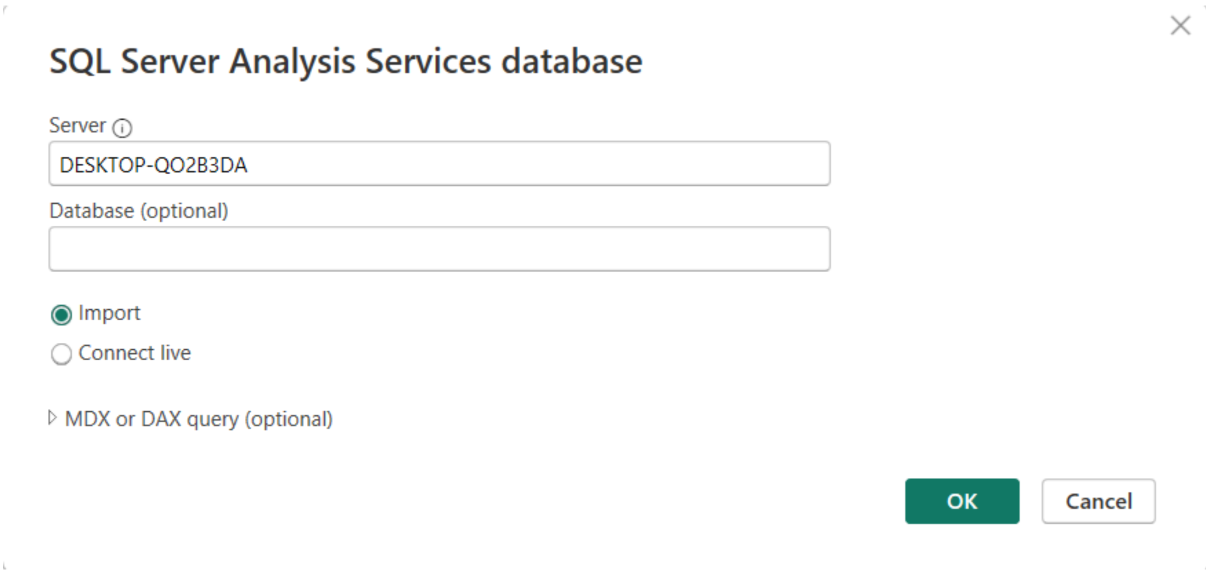
4.1.1 Getting Data from Analysis Services

Select Analysis Services from the “Get Data” option from Home tab



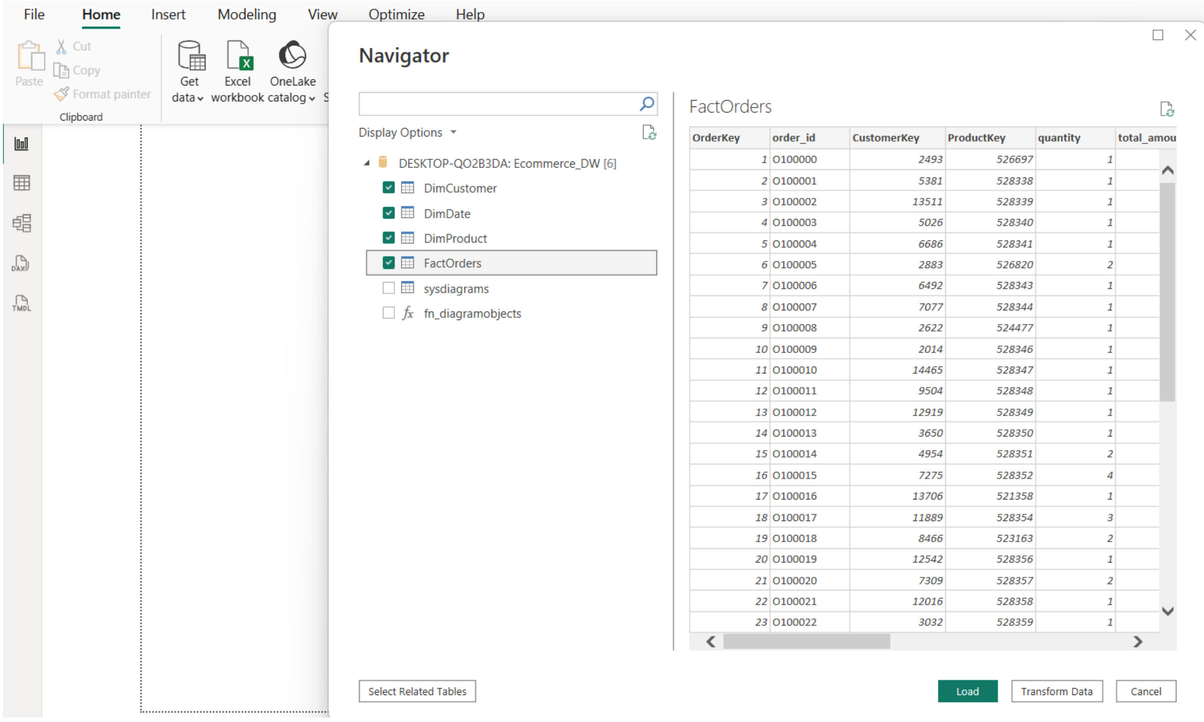
Figer-22

The server’s name was entered in the connection dialog, and Import was selected as the data connectivity mode, as the report was intended to be published to the Power BI Service. From the available options, the appropriate database Ecommerce_SSAS was entered to access the cube.



Figers-23

Once connected, the necessary dimensions and measures were chosen to form the Ecommerce_SSAS and imported into the Power BI model to be used in report creation.



OrderKey	order_id	CustomerKey	ProductKey	quantity	total_amou
1	O100000	2493	526697	1	
2	O100001	5381	528338	1	
3	O100002	13511	528339	1	
4	O100003	5026	528340	1	
5	O100004	6686	528341	1	
6	O100005	2883	526820	2	
7	O100006	6492	528343	1	
8	O100007	7077	528344	1	
9	O100008	2622	524477	1	
10	O100009	2014	528346	1	
11	O100010	14465	528347	1	
12	O100011	9504	528348	1	
13	O100012	12919	528349	1	
14	O100013	3650	528350	1	
15	O100014	4954	528351	2	
16	O100015	7275	528352	4	
17	O100016	13706	521358	1	
18	O100017	11889	528354	3	
19	O100018	8466	523163	2	
20	O100019	12542	528356	1	
21	O100020	7309	528357	2	
22	O100021	12016	528358	1	
23	O100022	3032	528359	1	

Figers-24

4.2 Report Demonstrations

Several reports were created to demonstrate the capabilities of Power BI. These include matrix reports, dashboards with slicers, drill-down reports, and drill-through reports. Each report provides a different way of analyzing data, helping users understand various aspects of business performance.

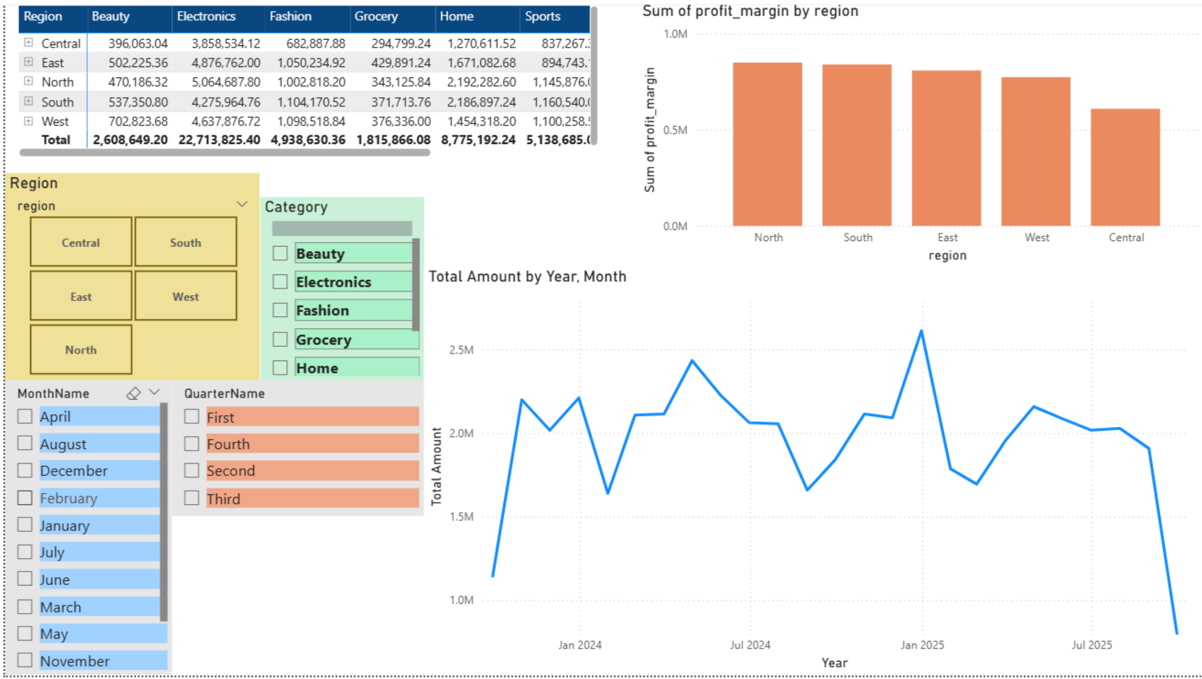
4.2.1 Report 1: Report with a matrix

A matrix is like a table, except it is set up to display data in columns and rows, with aggregate data at the intersections.

Region	Beauty	Electronics	Fashion	Grocery	Home	Sports	Toys	Total
Central	396,063.04	3,858,534.12	682,887.88	294,799.24	1,270,611.52	837,267.36	372,048.88	**7,712,212.04**
East	502,225.36	4,876,762.00	1,050,234.92	429,891.24	1,671,082.68	894,743.12	890,127.12	**10,315,066.44**
North	470,186.32	5,064,687.80	1,002,818.20	343,125.84	2,192,282.60	1,145,876.08	451,688.00	**10,670,664.84**
South	537,350.80	4,275,964.76	1,104,170.52	371,713.76	2,186,897.24	1,160,540.00	682,630.00	**10,319,267.08**
West	702,823.68	4,637,876.72	1,098,518.84	376,336.00	1,454,318.20	1,100,258.52	574,074.64	**9,944,206.60**
Total	**2,608,649.20**	**22,713,825.40**	**4,938,630.36**	**1,815,866.08**	**8,775,192.24**	**5,138,685.08**	**2,970,568.64**	**48,961,417.00**

Figer-25

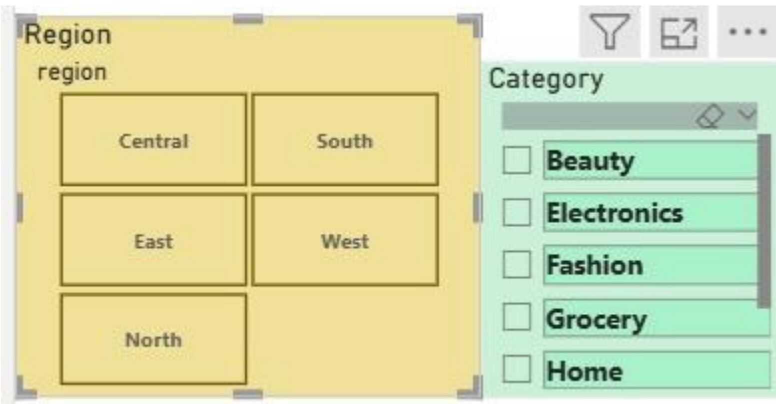
4.2.2 Report 2: Report with cascading slicers and multiple visuals



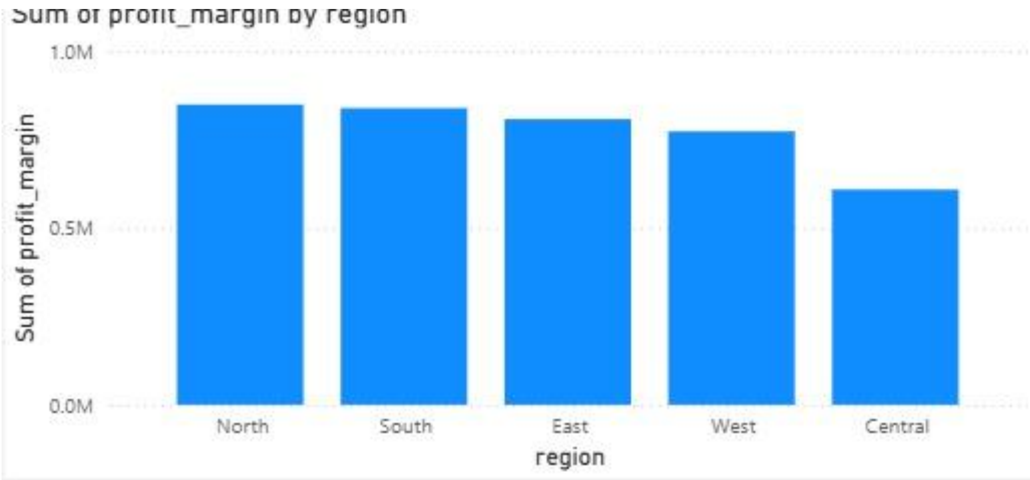
Figer-26

4.2.3 Report 3: Slice report

A report with Slice functionality was created to allow navigation from summary data to individual records.

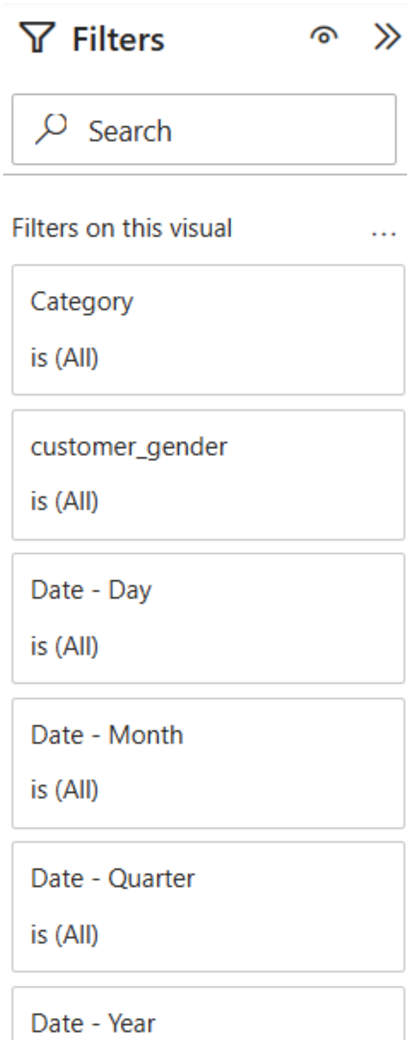


Figer-27



Figer-28

Drill-through Columns



Figer-29

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